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pyRTX: a Python package for high precision computation of non gravitational forces on deep space probes

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Summary

With the constant improvement of radiometric tracking systems, inaccuracies in the non-gravitational force modeling have become one of the limiting factors to deep space precise orbit determination, and the scientific products that it enables. The main factor impacting the limited accuracy of non-gravitational force models is the complex 3D shape of the spacecraft. While fast, reliable, analytical models are available for simple shapes (spheres, cubes, etc), no such model is generally available for a complex shape. This software package aims to address this limitation by legeraging ray-tracing to compute the complex interaction between the forcing environment (radiation, atmosphere) and the three dimensional shape of the spacecraft. This software is specifically geared towards planetary OD where inaccuracies of the model often cannot be overcome with continuous tracking as it is routinely done for Earth-orbiting spacecraft.

Statement of need

Several scientific investigations require high-precision reconstruction of spacecraft trajectories. Among these, one of the most demanding is the determination of the gravity field of Solar System bodies (planets, moons). This task is accomplished by solving the so-called orbit determination (OD) problem (Milani & Gronchi, 2009; Tapley et al., 2004). The solution of the OD in the adjustment of a dynamical model (a set of differential equations) describing the spacecraft motion. Systematic errors in the dynamical model will almost inevitably lead to systematic errors in the solution. In the recent years significant improvements in radiometric tracking system, have led to more and more precise measurements of the spacecraft position and velocity (the input to the OD), thus requiring increasingly more accurate dynamical models (Asmar et al., 2005; Cappuccio et al., 2020; Mazarico et al., 2023). One of the major limitations of current dynamical modelling of deep-space probes consists in the complex interaction between the spacecraft shape and the atmosphere, and with radiative forces (solar radiation pressure, albedo, thermal infrared radiation). We developed the pyRTX software package to address this limitation. Leveraging the ray-tracing technique, originally developed in computer graphics, instead of relying on simplified macro-models (flat plates, cylinders, etc) pyRTX is able to use the actual 3D shape of the spacecraft (provided as, for example, an .obj file), to compute several non-gravitational accelerations.

pyRTX addresses the gap in open source solutions for a comprehensive modelling of several, important, non-gravitational forces. Being built around the NAIF SPICE astrodynamic library (Acton, 1996) and its Python wrapper (Annex et al., 2020), pyRTX is intended to be a plug-in tool that can be used in existing OD codes, and applications.

Functionality

In this section we describe the main functionalities of the pyRTX software. All of these functionalities are discussed in the set of example scripts and notebooks included in the code distribution. Our companion paper (Zurria et al., 2026) discusses an actual application of the pyRTX library for ameliorating the OD of NASA's Lunar Reconnaissance Orbiter.

- *Solar Radiation Pressure Modeling*: pyRTX computes the acceleration due to solar photons by casting rays from a pixel plane representing the incoming solar flux. The engine inherently accounts for self-shadowing (where spacecraft components block light from reaching others) and multiple levels of reflections. Users can specify optical properties for each mesh face, allowing the software to simulate both specular and diffuse (Lambertian) reflections.
- *Planetary Radiation Pressure*: The software models the effects of radiation reflected (albedo) and emitted (thermal infrared) by planetary bodies. This includes the ability to use complex planetary shape models (e.g., topography from digital elevation models, DEMs) and spatially variable maps for albedo, emissivity, and surface temperature.
- *Eclipse and Shadow Function Analysis*: pyRTX can compute precise shadow functions during eclipse transitions. It supports advanced modeling features such as solar limb darkening, where the variation in intensity across the solar disk is accounted for in the flux calculation.
- *Atmospheric Drag*: For low-altitude missions, the software calculates the effective aerodynamic cross-section of the spacecraft, providing inputs for atmospheric drag modeling.
- *Lookup Table (LUT) Generation*: To enable efficient integration with OD software, pyRTX can pre-compute accelerations over a grid of incident directions. This feature supports spacecraft with articulating components (e.g., solar arrays), allowing users to generate comprehensive LUTs that map acceleration vectors to specific spacecraft orientations and strongly reduce computation time.

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